Addition and subtraction of algebraic terms

1 Simplify:

a $8x + 6y - 2y - 4x$

b $10m + 9n + 5m + 10n$

c $10x - 9y - 7x + 11y$

d $11m + 8n + 14m$

e $9xy + 12yx$

f $6p + 8q - 6p$

g $9p^2 - 11p + 11p^2$

h $7n^2 + n^2 - 4m + 2m^2 - 5n^2$

i $8x + 9y - 5x + 10y$

j $7a + 11a - 9b + b$

k $12x^2 + 10x - 9x - 5x^2$

l $8x - 7y - 6z + 10z$

m $15y - 8y^2 + 10y - y^2$

n $19a - 8b + 4c - 8a + b$

o $2ab - 7c + ab + 9c$

p $12j + 9k - 7k + 11j$

q $18x - 9y + 7x + 11y$

r $11y - 6z + y + 6z$

s $13m + 2n - 8m + 10n$

t $-3s + 4t - 6t + 9s$

2 Simplify:

a $5x + 2y - 8x + 6z - 3y + 4z$

b $2xy + 5yz + 4xz - 8xy + 2xz - 3yz$

c $8x^2 + 10x - 6 + 4x^2 - x + 11$

d $20 - 9x + 4y + 6y - 7x + 21$

e $9mn + 8m - 7n + 7n - 10m + 2mn$

f $-6 + 8x^2 - 9 + 5x + 16x^2$

g $11s + 6t + 3st - 2ts + t - 4s$

h $13ab - 8bc + 6ba - 9b + 11cb$

i $10x + 9y - 3y - 2x$

j $10p - 7q + 10 - 3p + 7q - 3$

k $10pq + 4p^2q - 8qp + 7p^2q$

l $20x^2 - 12x^2 + 10x^2 - x^2$

m $6zxy + 4xzy + 10xyz + 12xy + 11yx$

n $-2x + 2y - 6y - 10x$

o $-4rk + 3r - 6k - 2rk$

p $-6vw - 4v^2w + 2v^2w - 8wv$

q $-5m + n - (-7m) - 6n - 2m + 7$

r $-2m + n - (-6m) - 3n - 4m + 5$

Multiplication of algebraic terms

3 Simplify the following:

a $10 \times 6u$

b $7 \times (-3u)$

c $5x \times 2$

d $11 \times 3y$

e $(-8x) \times 9$

f $(-12) \times (-2u)$

g $(-3w) \times 2$

h $10 \times (-3y)$

4 Simplify the following:

a $x \times x$

b $2x \times 3x$

c $4y \times y$

d $3a \times 5a$

e $8x \times 2xy$

f $9b \times 3b$

g $(10a)^2$

h $(4x)^2$

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5 Simplify the following:

a $9 \times m \times n \times 8$

b $w \times 4 \times z \times 6$

c $10 \times (-r) \times s \times (-5)$

d $9r \times 6s$

e $6u^2 \times 7v^8$

f $16p^3 \times 14q^3$

g $(-2a) \times (-4b)$

h $3w \times (-7z)$

i $(-5r) \times (-4s)$

j $(-10r^8) \times 6s^7$

k $4p^5 \times (-3q^5)$

l $2a^4 \times (-5b^3)$

6 Simplify the following:

a $6r \times 2 \times 8s$

b $7w \times 9x \times 10y$

c $(-2w) \times (-4x) \times (-10y)$

d $(-8h) \times 5k \times (-3r) \times (-4s)$

e $5x \times 2y \times (-9)$

f $4a \times (-2a) \times 10a$

g $(-5x) \times (-2x) \times (-3y) \times (-3)$

h $(-2h) \times 4k \times (-j) \times (-5i)$

Division of algebraic terms

7 Simplify the following:

a $\frac{2x}{2}$

b $\frac{15v}{5}$

c $\frac{5m}{20}$

d $\frac{n}{4n}$

e $\frac{12xy}{12}$

f $\frac{63pq}{9p}$

g $\frac{12mn}{15m}$

h $\frac{6r}{rw}$

i $\frac{pr}{pqr}$

j $\frac{10u^6v^4}{u^6}$

k $\frac{-24a}{4}$

l $\frac{-11y}{y}$

m $\frac{y}{-11y}$

n $\frac{-12u}{3u}$

o $\frac{-4m}{-9m}$

p $\frac{-abc}{b}$

q $\frac{k}{-jk}$

r $\frac{-6j}{jk}$

s $\frac{-ac}{abc}$

t $\frac{-2b^3}{3b^3}$

u $\frac{-3r^3w^5}{r^3}$

v $\frac{12n}{n}$

8 Simplify the following:

a $5m \div 40$

b $20wz \div 4wz$

c $10r^6 \div 5r^6$

d $(-24r^4) \div 6r^4$

e $(-44rs) \div 4r$

f $(-36uv) \div (-6uv)$

g $10mn \div 5m$

h $18xy \div 6y$

i $27r^2 \div 9r$

j $(-20x^4) \div 10x^4$

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k $(-22x^2y) \div -2xy$

l $(-50abc) \div (-5ab)$

9 Simplify the following:

a $\dfrac{2x \times 3y}{xy}$

b $\dfrac{(-4x) \times 5y}{-10xy}$

c $\dfrac{6x \times 4xy}{8y}$

d $\dfrac{7x \times 4y}{2x \times y}$

e $\dfrac{10a \times 3b}{5b \times 2}$

f $\dfrac{12x \times 6y}{8x \times 2y}$

g $9x \times 4x \div 2x$

h $(-5x) \times 8y \div 10y$

i $11xy \times 3y \div y$

j $20y \div 4 \times 3y$

k $(10y \div 2y) \times (4x \div 2)$

l $(6x \times 3y) \div (9x \times 2)$

Applications

10 An isosceles triangle has 2 sides of length 5 cm. If the third side has a length of a cm, write an expression for the perimeter of the triangle.

11 Laura decides to put a bird feeder in her back garden. The first day she sees 6 birds use the feeder. The next day she sees 12 birds, and on the third day she sees 18 birds.

a If the number of birds continues to follow the pattern, calculate how many birds will Laura see on the fourth day.

b If the pattern continues, write an expression for the number of birds Laura will see after y days.

12 A tap has not been turned off properly, and water is dripping into the bucket underneath it. After 1 hour, the water level in the bucket is 3 cm. After 2 hours the water level is at 6 cm, and after 3 hours the water level is at 9 cm.

a If the tap is turned off after x hours, write an expression in terms of x for the water level in the bucket at this time.

b Describe the expression as it relates to the water level.

13 Tracy saves all of her 5 cent coins and 10 cent coins. After some time, she loses track of how many coins she has.
If p represents the number of 5 cent coins and q represents the number of 10 cent coins, write an expression for the total value of the coins that Tracy has.

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14 Uther likes to go kayaking, and on Saturday he takes his boat down to the lake which is 300 m away from his house. After 1 minute of paddling he is 370 m away from his house. After 2 minutes he is 440 m away, and after 3 minutes he reaches 510 m away.

a Uther paddles for t minutes in total before he changes direction. Write an expression for how far he is from his house at this time.

b Describe the expression as it relates to Uther's paddling pace.

15 To get to school in the morning, Sally walks for 7 minutes to the bus stop and then waits for 2 minutes for the bus to arrive. She then catches the bus the rest of the way to school.

a If the bus trip takes y minutes, write an expression in terms of y for the total length of time it takes Sally to get to school.

b Explain what the constant in your expression represents.

16 While Judy is packing rectangular boxes into crates, she notices that each crate is 12 times wider than the width of one box, and 11 times longer than the length of one box. Judy wants to know the greatest number of boxes she can pack into each crate.
Let the length of one box be L cm, and the width of one box be W cm.

a Find an expression for the volume of one box with a height of 44 cm.

b Find an expression for the volume of a crate of height H cm.

c Find an expression for the number of identical boxes that Judy can fit into each crate.

d If the crate is 88 cm high, calculate how many boxes can Judy fit into each crate.

17 Aaron have books to be placed on a shelf which is 6 times wider than the width of the book and 8 times longer than the length of the book. Aaron wants to know the greatest number of books he can place into the shelf.
Let the length of one book be x cm, and the width of one book be y cm.

a Find an expression for the volume of one book with a height of 6 cm.

b Find an expression for the volume of the shelf of thickness z cm.

c Find an expression for the number of books that Aaron can fit into the shelf.

d If the shelf is 15 cm thick, how many books can Aaron fit into the shelf?

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